

MATH 330

Linear Algebra

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Chapter 1

Vectors and Matrices

1.1 Euclidean Vectors and Linear Combinations

A **vector** in $\mathbb{R}^n$ is an ordered list of n real numbers. Geometrically, a vector represents a directed line segment possessing both magnitude and direction. Algebraically, it is represented as a column matrix.

Definition 1.tcb@cnt@definition: Euclidean Vector Space $\mathbb{R}^n$

For a positive integer n , the set of all n -tuples of real numbers is called **n -dimensional Euclidean space**, denoted by $\mathbb{R}^n$:

$$\mathbb{R}^n = \left\{ \underline{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} \mid v_1, v_2, \dots, v_n \in \mathbb{R} \right\}$$

The real numbers $v_1, v_2, \dots, v_n$ are called the **components** (or coordinates) of the vector $\underline{v}$.

Operations on Vectors

The algebraic structure of $\mathbb{R}^n$ is established by two fundamental operations: **vector addition** and **scalar multiplication**.

Definition 1.tcb@cnt@definition: Vector Addition

Let $\underline{u}, \underline{v} \in \mathbb{R}^n$ be vectors given by:

$$\underline{u} = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix}, \quad \underline{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$$

The **sum** $\underline{u} + \underline{v}$ is the vector in $\mathbb{R}^n$ obtained by adding corresponding components:

$$\underline{u} + \underline{v} = \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{pmatrix}$$

Example 1.tcb@cnt@example: Vector Addition

Let $\underline{u} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\underline{v} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$ in $\mathbb{R}^2$. Their sum is:

$$\underline{u} + \underline{v} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} 4 \\ 3 \end{pmatrix} = \begin{pmatrix} 2+4 \\ 1+3 \end{pmatrix} = \begin{pmatrix} 6 \\ 4 \end{pmatrix}$$

Definition 1.tcb@cnt@definition: Scalar Multiplication

Let $\underline{v} \in \mathbb{R}^n$ and let $c \in \mathbb{R}$ be a real number (called a **scalar**). The **scalar product** $c\underline{v}$ is obtained by multiplying each component of $\underline{v}$ by c :

$$c\underline{v} = c \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} = \begin{pmatrix} cv_1 \\ cv_2 \\ \vdots \\ cv_n \end{pmatrix}$$

Example 1.tcb@cnt@example: Scalar Multiplication

Let $c = -3$ and $\underline{v} = \begin{pmatrix} 1 \\ -2 \\ 5 \end{pmatrix} \in \mathbb{R}^3$. Then:

$$c\underline{v} = -3 \begin{pmatrix} 1 \\ -2 \\ 5 \end{pmatrix} = \begin{pmatrix} (-3)(1) \\ (-3)(-2) \\ (-3)(5) \end{pmatrix} = \begin{pmatrix} -3 \\ 6 \\ -15 \end{pmatrix}$$

Theorem 1.tcb@cnt@theorem: Algebraic Properties of $\mathbb{R}^n$

For all vectors $\underline{u}, \underline{v}, \underline{w} \in \mathbb{R}^n$ and all scalars $c, d \in \mathbb{R}$, vector addition and scalar multiplication satisfy the following properties:

- i. **Commutativity of Addition:** $\underline{u} + \underline{v} = \underline{v} + \underline{u}$
- ii. **Associativity of Addition:** $(\underline{u} + \underline{v}) + \underline{w} = \underline{u} + (\underline{v} + \underline{w})$
- iii. **Additive Identity:** There exists a zero vector $\underline{0} = (0 \ 0 \ \dots \ 0)^\top \in \mathbb{R}^n$ such that $\underline{v} + \underline{0} = \underline{v}$
- iv. **Additive Inverse:** For each $\underline{v} \in \mathbb{R}^n$, there exists a vector $-\underline{v} = (-1)\underline{v}$ such that

$$\underline{v} + (-\underline{v}) = \underline{0}$$

v. **Distributivity over Vector Addition:** $c(\underline{u} + \underline{v}) = c\underline{u} + c\underline{v}$

vi. **Distributivity over Scalar Addition:** $(c + d)\underline{v} = c\underline{v} + d\underline{v}$

vii. **Associativity of Scalar Multiplication:** $c(d\underline{v}) = (cd)\underline{v}$

viii. **Scalar Identity:** $1\underline{v} = \underline{v}$

1.1.1 Linear Combinations and Span

Definition 1.1.1: Linear Combination

Let $\underline{v}_1, \underline{v}_2, \dots, \underline{v}_k \in \mathbb{R}^n$ be vectors and $c_1, c_2, \dots, c_k \in \mathbb{R}$ be scalars. A vector $\underline{w} \in \mathbb{R}^n$ given by

$$\underline{w} = c_1\underline{v}_1 + c_2\underline{v}_2 + \dots + c_k\underline{v}_k = \sum_{i=1}^k c_i\underline{v}_i$$

is called a **linear combination** of the vectors $\underline{v}_1, \underline{v}_2, \dots, \underline{v}_k$ with weights $c_1, c_2, \dots, c_k$.

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Exercise 1.1.1: Linear Combination

Find $c, d \in \mathbb{R}$ such that

$$c \begin{pmatrix} 2 \\ 1 \end{pmatrix} + d \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 8 \\ 2 \end{pmatrix}$$

Solution. Expanding the linear combination into component equations:

$$\begin{aligned} c \begin{pmatrix} 2 \\ 1 \end{pmatrix} + d \begin{pmatrix} 2 \\ -1 \end{pmatrix} &= \begin{pmatrix} 8 \\ 2 \end{pmatrix} \\ \begin{pmatrix} 2c \\ c \end{pmatrix} + \begin{pmatrix} 2d \\ -d \end{pmatrix} &= \begin{pmatrix} 8 \\ 2 \end{pmatrix} \\ \begin{pmatrix} 2c + 2d \\ c - d \end{pmatrix} &= \begin{pmatrix} 8 \\ 2 \end{pmatrix} \end{aligned}$$

This results in the linear system:

$$\begin{aligned} 2c + 2d &= 8 \\ c - d &= 2 \end{aligned}$$

Adding the second equation multiplied by 2 to the first gives $4c = 12$, so $c = 3$. Substituting $c = 3$ into $c - d = 2$ yields $3 - d = 2 \implies d = 1$. $\square$

Exercise 1.tcb@cnt@exercise: Describing All Linear Combinations

Describe all linear combinations of:

$$1. \underline{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad \underline{b} = \begin{pmatrix} 3 \\ 6 \\ 9 \end{pmatrix}$$

$$2. \underline{a} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \underline{b} = \begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix}$$

Solution.

1. Since $\underline{b} = 3\underline{a}$, the vectors $\underline{a}$ and $\underline{b}$ are collinear. Any linear combination $c\underline{a} + d\underline{b} = (c + 3d)\underline{a}$ is a scalar multiple of $\underline{a}$, which describes all vectors lying on the line through the origin in direction $(1 \ 2 \ 3)^\top$.
2. Since $\underline{a}$ and $\underline{b}$ are not parallel, their linear combinations span a 2-dimensional plane in $\mathbb{R}^3$ passing through the origin and containing both vectors.

□

1.2 Length and Angles from Dot Product

The dot product introduces metric geometry into Euclidean space $\mathbb{R}^n$, enabling the measurement of lengths, distances, and angles.

Definition 1.tcb@cnt@definition: Dot Product

Let $\underline{a}, \underline{b} \in \mathbb{R}^n$. The **dot product** (or **Euclidean inner product**) $\underline{a} \cdot \underline{b}$ is the scalar defined by:

$$\underline{a} \cdot \underline{b} := \underline{a}^\top \underline{b} = \sum_{i=1}^n a_i b_i = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n \in \mathbb{R}$$

Theorem 1.tcb@cnt@theorem: Properties of the Dot Product

For all vectors $\underline{a}, \underline{b}, \underline{c} \in \mathbb{R}^n$ and any scalar $k \in \mathbb{R}$:

- Symmetry:** $\underline{a} \cdot \underline{b} = \underline{b} \cdot \underline{a}$
- Distributivity:** $(\underline{a} + \underline{b}) \cdot \underline{c} = \underline{a} \cdot \underline{c} + \underline{b} \cdot \underline{c}$
- Homogeneity:** $(k\underline{a}) \cdot \underline{b} = k(\underline{a} \cdot \underline{b})$
- Positive Definiteness:** $\underline{a} \cdot \underline{a} \geq 0$, with $\underline{a} \cdot \underline{a} = 0$ if and only if $\underline{a} = \underline{0}$.

1.2.1 Vector Length and Norm

Definition 1.tcb@cnt@definition: Length / Norm

Let $\underline{a} \in \mathbb{R}^n$. The **length** (or **Euclidean norm**) of $\underline{a}$, denoted $\|\underline{a}\|$, is:

$$\|\underline{a}\| := \sqrt{\underline{a} \cdot \underline{a}} = \sqrt{a_1^2 + a_2^2 + \cdots + a_n^2}$$

Note that $\|\underline{a}\|^2 = \underline{a} \cdot \underline{a}$.

Definition 1.tcb@cnt@definition: Unit Vector

A vector $\underline{u} \in \mathbb{R}^n$ is called a **unit vector** if $\|\underline{u}\| = 1$.

Given any non-zero vector $\underline{v} \in \mathbb{R}^n$, dividing by its length yields a unit vector $\underline{u} = \frac{\underline{v}}{\|\underline{v}\|}$ in the direction of $\underline{v}$ (a process called **normalization**).

In $\mathbb{R}^2$, any unit vector extending from the origin to the unit circle can be parameterized by an angle θ as:

$$\underline{u} = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}$$

1.2.2 Orthogonality and Angles

Definition 1.tcb@cnt@definition: Orthogonal Vectors

Two vectors $\underline{a}, \underline{b} \in \mathbb{R}^n$ are **orthogonal** (denoted $\underline{a} \perp \underline{b}$) if their dot product is zero:

$$\underline{a} \cdot \underline{b} = 0$$

Orthogonal vs. Perpendicular The terms ‘orthogonal’ and ‘perpendicular’ have the same geometric idea. However, ‘orthogonal’ applies to vectors (and includes the zero vector $\underline{0}$, which is orthogonal to every vector), while ‘perpendicular’ is usually for geometric lines and planes.

Example 1.tcb@cnt@example: Pythagorean Theorem for Vectors

Let $\underline{a}, \underline{b} \in \mathbb{R}^n$ be orthogonal vectors ($\underline{a} \cdot \underline{b} = 0$). Then:

$$\|\underline{a} + \underline{b}\|^2 = \|\underline{a}\|^2 + \|\underline{b}\|^2$$

Proof. Expanding the squared norm using dot product properties:

$$\begin{aligned} \|\underline{a} + \underline{b}\|^2 &= (\underline{a} + \underline{b}) \cdot (\underline{a} + \underline{b}) \\ &= \underline{a} \cdot \underline{a} + 2(\underline{a} \cdot \underline{b}) + \underline{b} \cdot \underline{b} \\ &= \|\underline{a}\|^2 + 0 + \|\underline{b}\|^2 \\ &= \|\underline{a}\|^2 + \|\underline{b}\|^2 \end{aligned}$$

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Exercise 1.tcb@cnt@exercise: Dot Product

Let $\underline{a} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$ and $\underline{b} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$. Find $\underline{a} \cdot \underline{b}$.

Solution. Calculating the dot product:

$$\underline{a} \cdot \underline{b} = (4)(-1) + (2)(2) = -4 + 4 = 0$$

Since $\underline{a} \cdot \underline{b} = 0$, the vectors $\underline{a}$ and $\underline{b}$ are orthogonal. □

Definition 1.tcb@cnt@definition: Angle between Vectors

For non-zero vectors $\underline{a}, \underline{b} \in \mathbb{R}^n$, the **angle** $\alpha \in [0, \pi]$ between $\underline{a}$ and $\underline{b}$ satisfies:

$$\cos \alpha = \frac{\underline{a} \cdot \underline{b}}{\|\underline{a}\| \|\underline{b}\|}$$

$$\underline{a} \cdot \underline{b} = \|\underline{a}\| \|\underline{b}\| \cos \alpha.$$

Theorem 1.tcb@cnt@theorem: Vector Inequalities

For any vectors $\underline{a}, \underline{b} \in \mathbb{R}^n$:

1. Cauchy-Schwarz Inequality:

$$|\underline{a} \cdot \underline{b}| \leq \|\underline{a}\| \|\underline{b}\|$$

Equality holds if and only if $\underline{a}$ and $\underline{b}$ are linearly dependent (collinear).

2. Triangle Inequality:

$$\|\underline{a} + \underline{b}\| \leq \|\underline{a}\| + \|\underline{b}\|$$

Equality holds if and only if one vector is a non-negative scalar multiple of the other.

1.3 Intro to Matrices and Column Spaces

Matrices provide a structured representation for systems of linear equations, linear transformations, and collections of vectors.

Definition 1.tcb@cnt@definition: Matrix

An $m \times n$ **matrix** A is a rectangular array of real numbers formatted with m rows and n columns:

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} = [a_{ij}]_{m \times n}$$

Expressed in terms of its column vectors $\underline{a}_1, \underline{a}_2, \dots, \underline{a}_n \in \mathbb{R}^m$, A is written as:

$$A = (\underline{a}_1 \quad \underline{a}_2 \quad \cdots \quad \underline{a}_n)$$

1.3.1 Special Types of Matrices

Matrices possess structural classifications according to their dimensions and internal symmetry.

Definition 1.tcb@cnt@definition: Matrix Classifications

Let $A = [a_{ij}]$ be a matrix.

- **Square Matrix:** A matrix with equal numbers of rows and columns ($m = n$).
- **Symmetric Matrix:** A square matrix satisfying $A = A^\top$, meaning $a_{ij} = a_{ji}$ for all i, j .
- **Diagonal Matrix:** A square matrix where all off-diagonal entries are zero ($a_{ij} = 0$ for all $i \neq j$).
- **Identity Matrix (I_n):** A diagonal matrix with ones on the main diagonal ($a_{ii} = 1$) and zeros elsewhere.
- **Triangular Matrices:**
 - **Upper Triangular Matrix:** A square matrix with zeros below the main diagonal ($a_{ij} = 0$ for $i > j$).
 - **Lower Triangular Matrix:** A square matrix with zeros above the main diagonal ($a_{ij} = 0$ for $i < j$).

Example 1.tcb@cnt@example: Types of Matrices

- **3×3 Identity Matrix (Symmetric and Diagonal):**

$$I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- **3×3 Diagonal Matrix:**

$$D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

- **Triangular Matrices:**

- 2×2 Upper Triangular Matrix:

$$U = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$$

- 3×3 Lower Triangular Matrix:

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 3 & 1 & 5 \end{pmatrix}$$

1.3.2 Matrix-Vector Product**Definition 1.tcb@cnt@definition: Matrix-Vector Product**

Let $A = [a_{ij}]$ be an $m \times n$ matrix with rows $\underline{r}_1^\top, \underline{r}_2^\top, \dots, \underline{r}_m^\top \in \mathbb{R}^n$ and column vectors $\underline{a}_1, \underline{a}_2, \dots, \underline{a}_n \in \mathbb{R}^m$. For a vector $\underline{x} = (x_1 \ x_2 \ \dots \ x_n)^\top \in \mathbb{R}^n$, the **matrix-vector product** $A\underline{x} \in \mathbb{R}^m$ is defined componentwise by:

$$A\underline{x} = \begin{pmatrix} \underline{r}_1 \cdot \underline{x} \\ \underline{r}_2 \cdot \underline{x} \\ \vdots \\ \underline{r}_m \cdot \underline{x} \end{pmatrix}$$

Linear Combination Viewpoint Equivalently, the product $A\underline{x}$ is a **linear combination of the column vectors of A** with weights given by the entries of $\underline{x}$:

$$A\underline{x} = x_1 \underline{a}_1 + x_2 \underline{a}_2 + \dots + x_n \underline{a}_n$$

This dual perspective links matrix multiplication directly to linear combinations and vector spans.

Exercise 1.tcb@cnt@exercise: Matrix-Vector Product

Let $A = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$ and $\underline{h} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$. Find $A\underline{h}$.

Solution. Using the row dot-product formula:

$$\begin{aligned} A\underline{h} &= \begin{pmatrix} (1 \ 3) \cdot (1 \ 2)^\top \\ (2 \ 4) \cdot (1 \ 2)^\top \end{pmatrix} \\ &= \begin{pmatrix} (1)(1) + (3)(2) \\ (2)(1) + (4)(2) \end{pmatrix} = \begin{pmatrix} 7 \\ 10 \end{pmatrix} \end{aligned}$$

Alternatively, using column linear combinations:

$$A\underline{h} = 1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + 2 \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 6 \\ 8 \end{pmatrix} = \begin{pmatrix} 7 \\ 10 \end{pmatrix}$$

□

1.3.3 Column Space of a Matrix